CCIThink

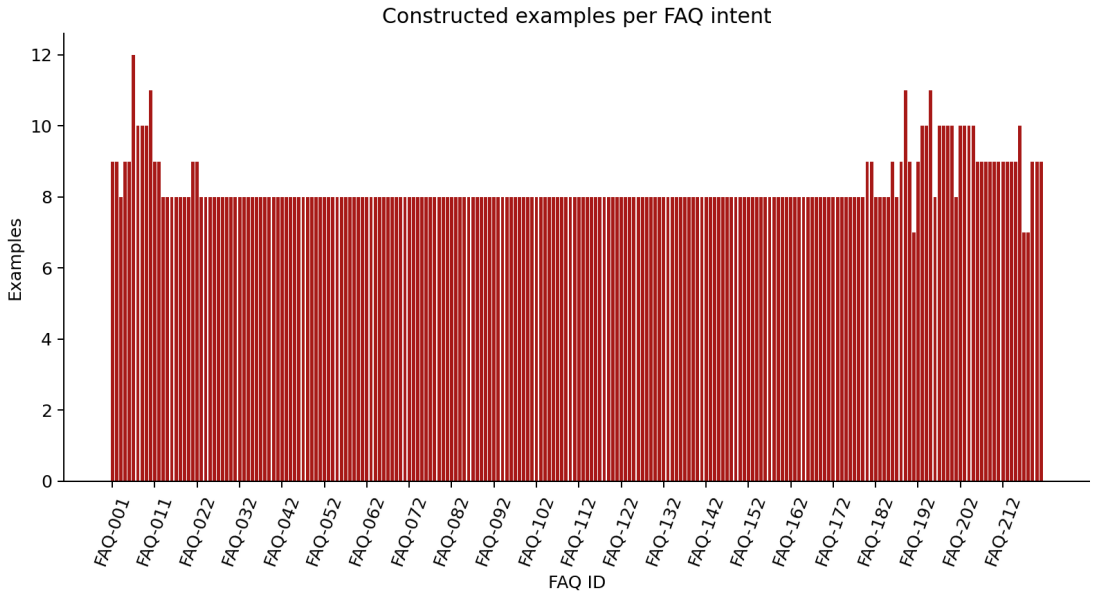
- CCIT information chatbot and explainable FAQ classifier
- Boncajes, Rhian Jared D.
- Marcos, Maricar N.
- Carandang, Jamie Pearl U.
- Principles of Artificial Intelligence and Data Science

Problem and objectives

- Find supported answers within 220 active department FAQs.
- Prepare data, describe patterns, train a classifier, and evaluate results.
- Keep unclear and unsupported questions separate from supported answers.

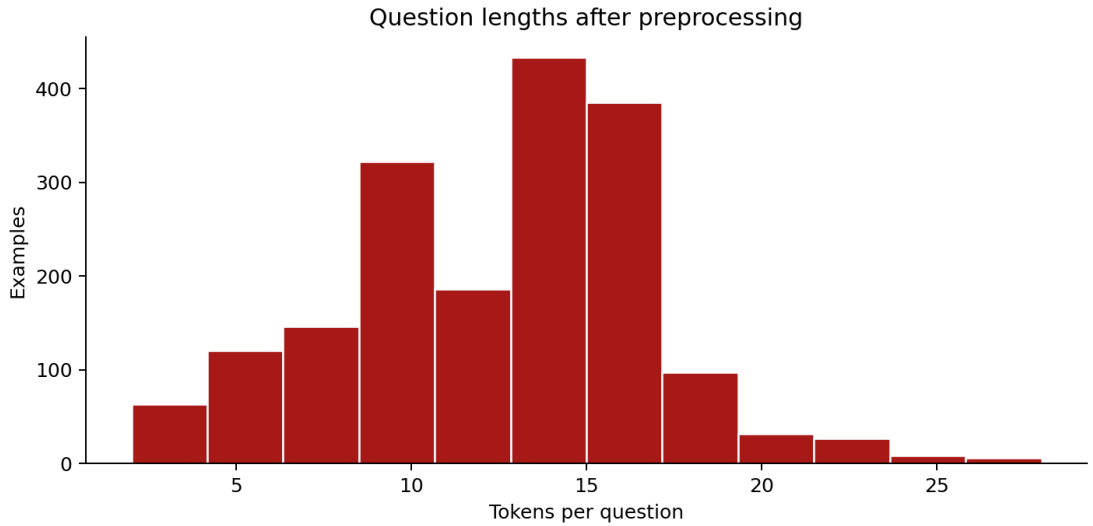
Dataset and source

- 1,825 labeled English questions; 220 FAQ labels.
- 945 train / 440 validation / 440 test.
- Training titles and variants were supplied with the knowledge.
- Holdouts are AI-authored constructed examples, not student transcripts.



Cleaning and exploration

- Check missing fields, duplicates, labels, and formats.
- Reject label conflicts and cross-split duplicates.
- 0 missing, 0 duplicate, 0 invalid rows removed.
- Mean 12.33 tokens; 9 long valid questions retained.



Coverage and bias

- All current dataset examples are English.
- The model test does not establish Filipino or Taglish accuracy.
- Topic counts are designed examples, not measured student demand.

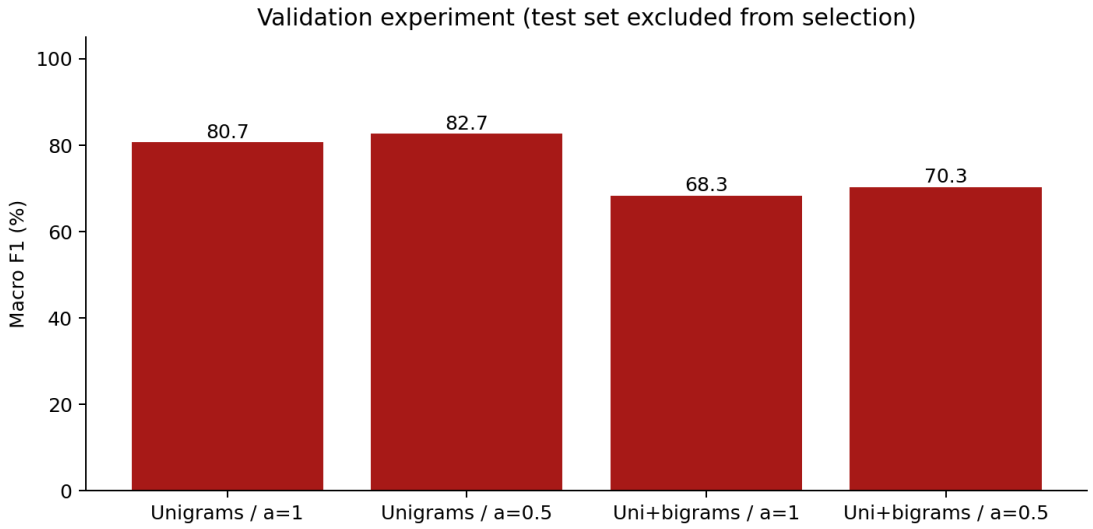


How Naive Bayes works

- Count words in training questions for each FAQ label.
- Estimate a label prior and smoothed word likelihoods.
- Add log likelihoods to rank possible labels.
- Independence is an approximation; scores are uncalibrated.

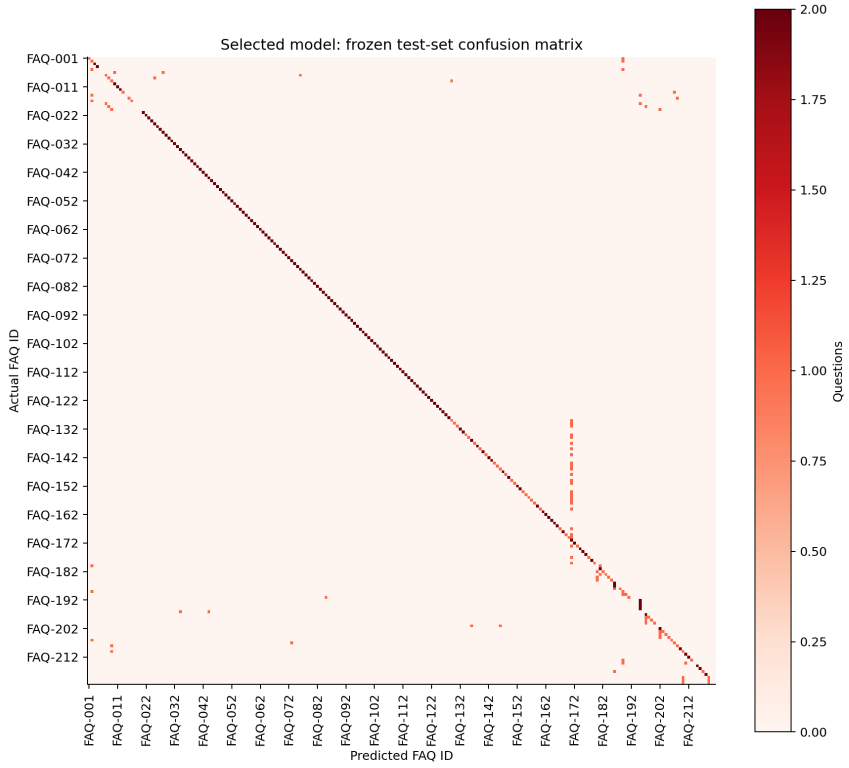
Parameter experiment

- Compare four fixed unigram/bigram and alpha settings.
- Select with validation macro F1; keep test questions separate.
- Selected: Unigrams / a=0.5.
- Validation: 366/440 correct; macro F1 0.827.



Test results

- 353/440 correct (80.2%).
- Macro F1 0.807.
- Majority baseline: 0.5% accuracy.
- 87 errors; two test questions per label.



Errors and limitations

- Similar course and curriculum wording can confuse labels.
- AI-authored questions may share templates.
- FAQ-020 was retired by user request; other IDs remain stable.
- Recomputed results do not constitute a new independent study.

Working demonstration

- Run python analyze.py, then start python app.py in FAQ mode.
- Use /analysis for five charts, metrics, and a local prediction.
- Use chat for source-supported answers and short forms.
- Try an ambiguous question and an unsupported detail.

Ethics and responsible AI

- No private student records are required.
- Source currency and permissions still need review.
- AI assistance is disclosed; authors must explain their work.
- Optional provider wording is not a verified live-generation result.

Conclusion and next steps

- A complete data-science workflow supports the chatbot.
- The classifier remains limited to a constructed, closed-set task.
- Next: confirmed sources, permissioned student questions, independent labels.
- Questions and defense discussion